12th HACKERS KA JUGAAD

Human health and Disease

1. As per good humor, hypothesis arrived at by reflective thought and asserted by hypocrates along with Indian Ayurveda system. Health is a state of body and mind where there was a balance of certain humors. Example persons with black bile belonged to hot personality and had fevers

2. William Harvey disproved this good humour, hypothesis of health by demonstrating normal body temperature in persons with black bile by using thermometer.

3. Mind influences through neural and endocrine systems and that are immune system maintains our health that is the state of complete physical, mental and social and psychological well-being

4. Health is not simply absence of disease or physical fitness

factors affecting health:

mental state, genetic disorders, infections, and life style (habits, rest and exercise)

Healthy conditions
- Increase → productivity, longevity
- Increase → bring economic prosperity
- Decrease → Infant and maternal mortality.

Disease — It's state of body when functioning of one or more organ/systems is adversely affected, characterised by various signs and symptoms.

Parameters
- transmission from one person to another
- example

Types of disease

	Non-infectious	infectious
	✗	✓
	cancer	AIDS

Pathogens are disease causing organisms

- most parasites are pathogens living in (or on) the host multiply and interfere with normal vital activities resulting in morphological and functional damage.

- gut pathogen can survive harsh pH and digestive enzymes.

Mode of transmission	Bacterial	Viral	Protozoan	Helminthic
Air (droplet/aerosol) or object borne (pens, knobs etc.)	Pneumonia, diphtheria	Common cold, Smallpox	-	-
Direct contact	Tetanus	Smallpox	-	-
Contaminated food and water	Typhoid, dysentery	Polio	Amoebiasis	Ascariasis
Insect vector/vector borne	Plague	Chikungunya, Dengue	Malaria	Filariasis
Body fluids	Syphilis	AIDS	Trichomoniasis	-

• **Vector:** Transmits disease from one organism to another e.g. female Aedes mosquito is the vector for dengue and chikungunya, while Anopheles spreads malaria.

Measures for preventing spread of infectious diseases

Parameters	Measures
Personal Hygiene	• Keeping the body clean • Consumption of clean drinking water, food, vegetables, fruits etc.
Public Hygiene	• Proper disposal of waste and excreta • Periodic cleaning and disinfection of water reservoirs, pools, cesspools and tanks. • Decontamination of drinking water
Avoid close contact	• Contact with infected persons and belongings should be avoided
Control vectors and their breeding places	• Avoid stagnation of water in and around residential areas. • Regular cleaning of house old coolers • Use of mosquito nets • Introducing larvicidal fishes like Gambusia in ponds that feed on mosquito larvae • Spraying of insecticides in ditches, drainage areas and swamps • Doors and windows should be provided with wire mesh.

Bacterial diseases

Disease	Pathogen	Organ affected	Common symptoms
• Typhoid	Salmonella typhi Diagnostic test: Widal test	Small intestine and other organs by migrating through blood	• Sustained high fever (39-40ºC) • Stomach pain • Weakness • Constipation • Headache • Loss of appetite • In severe cases, intestinal perforation and death may occur.
• Pneumonia	Streptococcus pneumoniae, Haemophilus influenzae	Alveoli of lungs (fluid)	• Problem in respiration due to fluid filled alveoli • Fever, chills, cough, headache • In severe cases, lips and finger nails turn gray to bluish

Viral diseases

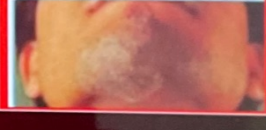
Disease	Pathogen	Organ affected	Symptoms
• Common cold	Rhino virus	Nose and respiratory passage	• Nasal congestion and discharge • Sore throat • Hoarseness, cough • Headache, tiredness

• Common cold does not infect lungs and its symptoms usually lasts for 3-7 days

Helminthic diseases

Disease	Pathogen	Organ/structure affected	Symptoms	
• Ascariasis	Ascaris (Roundworm)	Intestine	• Internal bleeding, fever, muscular pain, anemia, blockage of intestinal passage	
•Elephantiasis /Filariasis	Wuchereria bancrofti /W. malayi (Filarial worm)	Lymphatic vessels	• Chronic inflammation of organs in which they live for many years resulting in gross deformities e.g., limbs, genital organs etc.	

Fungal diseases

Disease	Pathogen	Body parts affected	Symptoms	
• Ringworm	Microsporum, Trichophyton, Epidermophyton	Skin, nails, scalp	• Dry, scaly lesions • Intense itching	

• **Heat and moisture** makes the fungi thrive in **skin folds** such as in groin and between toes
• Acquired from soil or belongings of infected individuals such as towels, combs, clothes etc.

Protozoal diseases

Disease	Pathogen	Area affected	Symptoms
• Amoebiasis /Amoebic dysentery	Entamoeba histolytica	Large Intestine	• Constipation • Abdominal pain • Cramps • Stool with excess mucous and blood clots
• Malaria	• Plasmodium • P. vivax • P. malariae • P. Falciparum (most fatal)	RBCs	• Chills • High fever recurring every 3-4 days • If not treated, can prove to be fatal

• House flies act as **mechanical carrier for amoebiasis**
• *P. falciparum* causes **malignant malaria (Most serious form)**

Malaria

Host < primary host → ♀ anopheles mosq. (sexual cycle)
secondary host → human (asexual cycle)

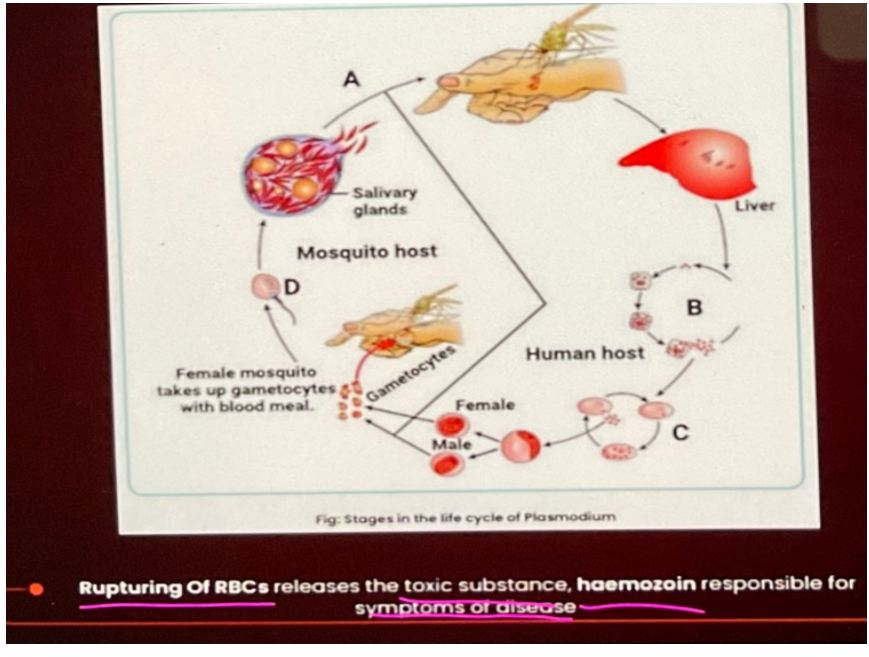


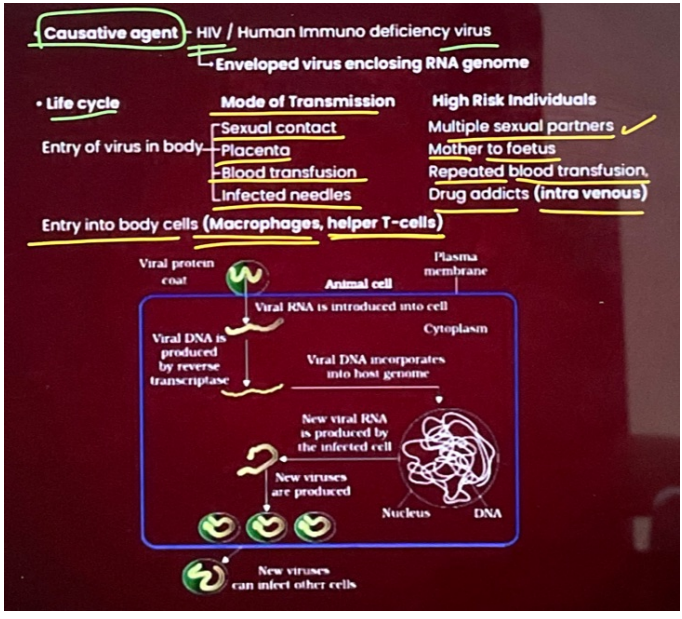
Fig: Stages in the life cycle of Plasmodium

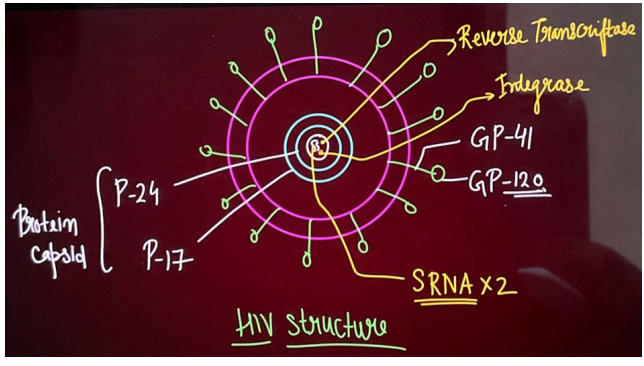
Rupturing Of RBCs releases the toxic substance, **haemozoin** responsible for symptoms of disease

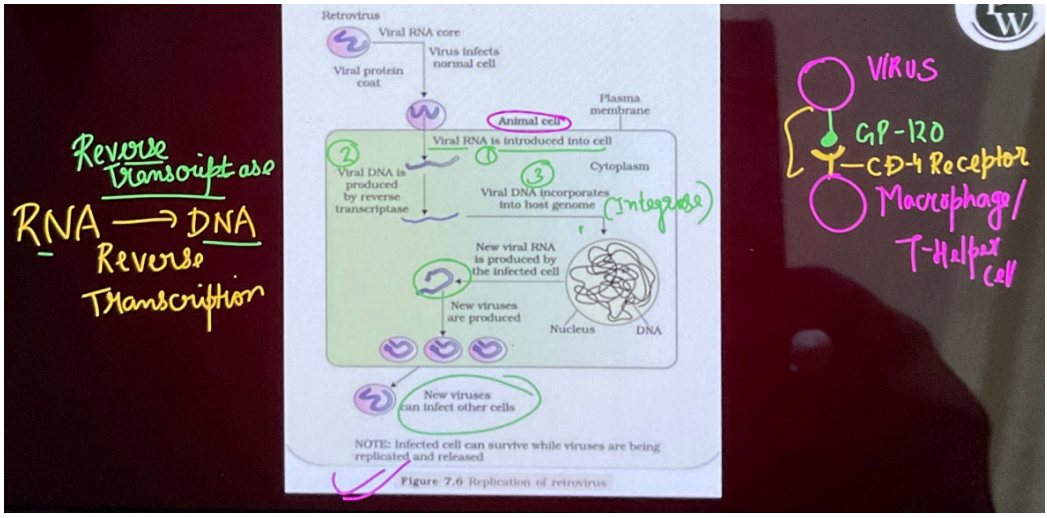
AIDS/ acquired immuno deficiency Syndromes

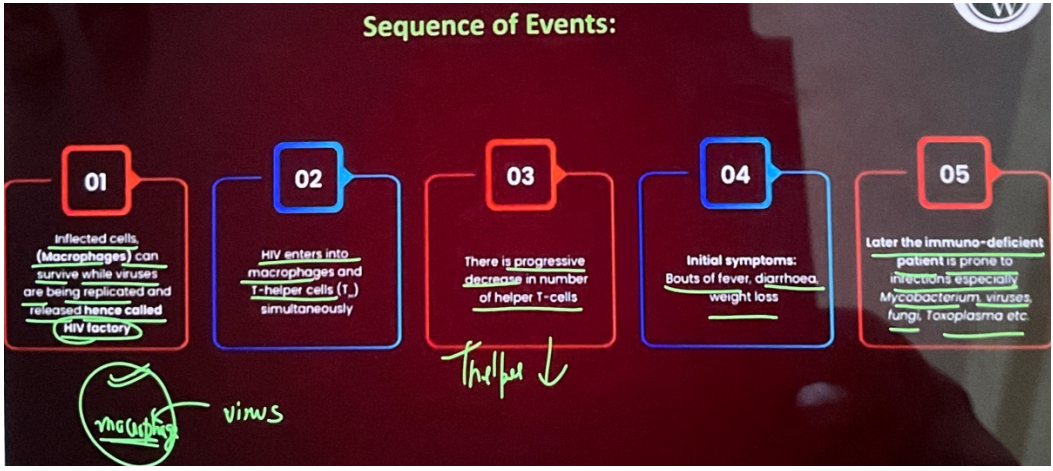
- 1st reported – 1981, Killed – approximately 25 million people in last 25 years.

- It's deficiency of immune system, acquired during the lifetime of an individual.
- Syndrome means group of symptoms.
- non congenital, fatal infectious disease









There is always a time-lag between infection and appearance of AIDS symptoms. This may vary from a few months to many years. (usually 5-10 yrs).

ELISA (enzyme linked immuno sorbent Assay) ← Diagnostic test

Treatment → anti-retroviral drugs, can only prolong life but cannot prevent death.

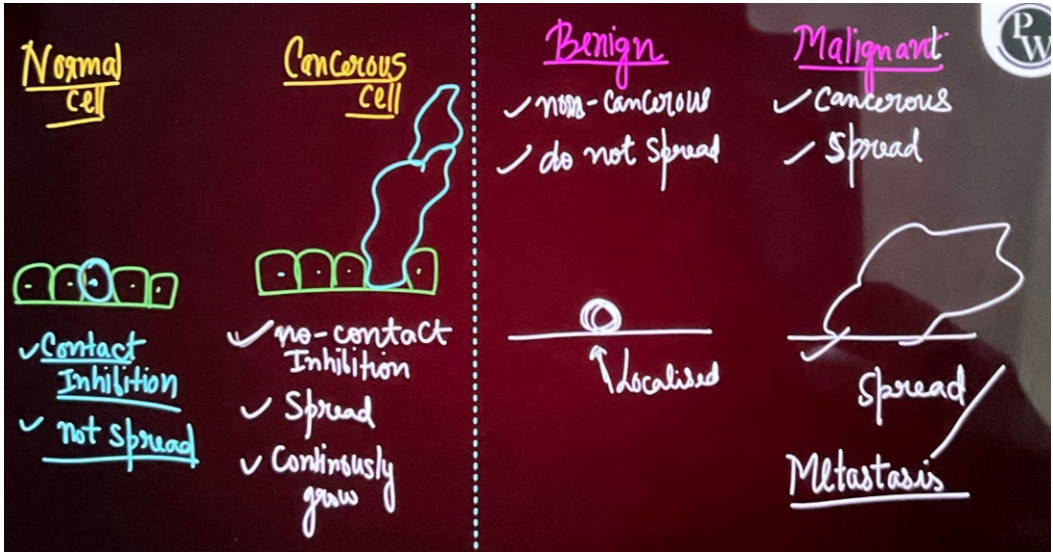
Cancer

- A dreaded non-infectious disease; major cause of death all across the globe.

Parameters	Normal cells	Cancerous cells/Neoplastic cells
Cell growth and differentiation	Highly controlled and regulated.	Uncontrolled & non-regulated.
Contact inhibition	**Present**, virtue of which contact with other cells inhibits their growth.	**Lost**, so these cells keep on dividing and form mass of cells called Tumor/Neoplasm.

Types of Tumor

Parameters	Benign	Malignant tumor/cancer
Cell growth and differentiation	Confined to original place	Grow rapidly and spread to other parts.
Contact inhibition	Little damage.	Invade and damage other cells starving normal cells by competing for vital nutrients.
Metastasis	No	**Yes**, Cells sloughed from such tumors reach distant sites through blood and start new tumor called **Metastasis (most feared property)**.



Causes/Agents

normal proto/cellular proto oncogene → neoplastic transforming

↓
Cancerous cell
oncogene

- ionising radiations
  - X-rays, Y-rays
- non-ionizing radiations - UV rays
- Physical agents
- chemical agents - in tobacco smoke
- Biological agents - oncogenic viruses (carry viral oncogenes)
- Diagnosis/ detection - early detection allows the disease to be treated successfully in many cases.

Diagnosis of Cancer

Technique	Basis	Detect
Biopsy	Histopathological studies	Changes in tissue
Blood and bone marrow test	Cell counts	Leukemias
Radiography	X-rays	Internal organ cancers
Computed tomography (CT) (3D)	X-rays	Internal organ cancers (3D image)
Magnetic resonance Imaging (MRI)	Strong magnetic fields and non-ionising radiations	Accurately detect pathological and physiological changes in living tissue
Molecular techniques	Identification of genes responsible for susceptibility to certain cancers	
Antibodies based	Against cancer specific antigens	Certain cancers

Treatment of Cancer

- Tumor cells have ability to avoid detection and destruction by immune system.

- Approaches for treatment – surgery
  - Radiotherapy: tumor cells irradiated lethally.
  - chemotherapy: side effects like hair loss, anemia
  - Immunotherapy (α- interferons) (Biological response modifiers) activate immune system and helps in destroying the tumor.

Immunity

The ability of the host to fight the disease causing organisms, conferred by the Immune system is called immunity.

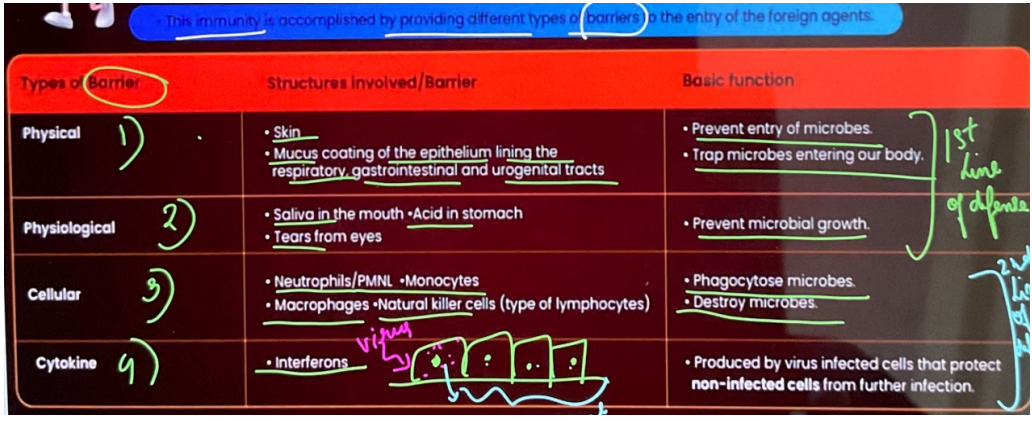
Parametes observed from exposure to infection
Defence
Memory record.

Types

Innate	Acquired.
– time of birth	– after birth
– not required	– required
– non specific	– specific.

Memory based immunity evolved in higher Vertebrates.

Innate Immunity



This immunity is accomplished by providing different types of barriers to the entry of the foreign agents.

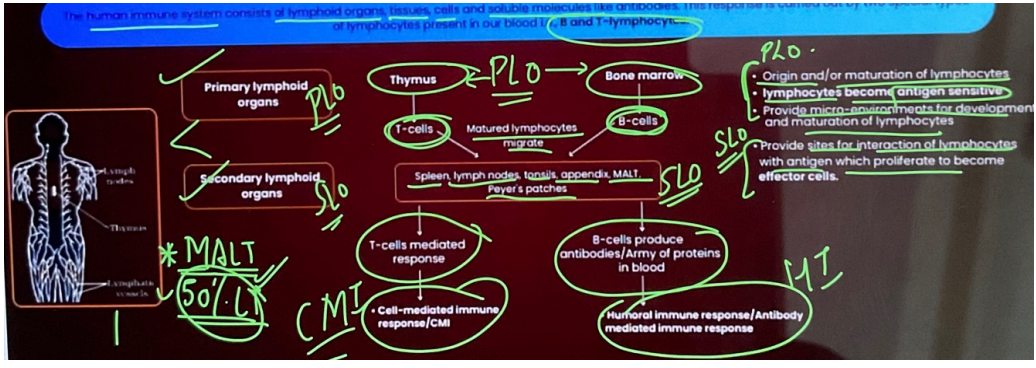
Types of Barrier	Structures involved/Barrier	Basic function
Physical 1)	• Skin • Mucus coating of the epithelium lining the respiratory, gastrointestinal and urogenital tracts	• Prevent entry of microbes. • Trap microbes entering our body.
Physiological 2)	• Saliva in the mouth • Acid in stomach • Tears from eyes	• Prevent microbial growth.
Cellular 3)	• Neutrophils/PMNL • Monocytes • Macrophages • Natural killer cells (type of lymphocytes)	• Phagocytose microbes. • Destroy microbes.
Cytokine 4)	• Interferons	• Produced by virus infected cells that protect **non-infected cells** from further infection.

1st line of defense

2nd line of defense

virus

Acquired immunity



The human immune system consists of lymphoid organs, tissues, cells and soluble molecules like antibodies. This response is carried out by the specialized types of lymphocytes present in our blood i.e. B and T-lymphocytes.

Primary lymphoid organs

PLO

Thymus

Bone marrow

T-cells

Matured lymphocytes migrate

B-cells

Secondary lymphoid organs

SLO

Spleen, lymph nodes, tonsils, appendix, MALT, Peyer's patches

T-cells mediated response

B-cells produce antibodies/Army of proteins in blood

• Cell-mediated immune response/CMI

Humoral immune response/Antibody mediated immune response

CMI

HI

* MALT

50% of

PLO.

- Origin and/or maturation of lymphocytes
- lymphocytes become antigen sensitive
- Provide micro-environments for development and maturation of lymphocytes
- Provide sites for interaction of lymphocytes with antigen which proliferate to become effector cells.

SLO

- Type
- exposure to pathogens

primary immune response
first time

↓ subsequent times.

secondary immune response

Intensity

low → high

Based on memory of 1st encounter.

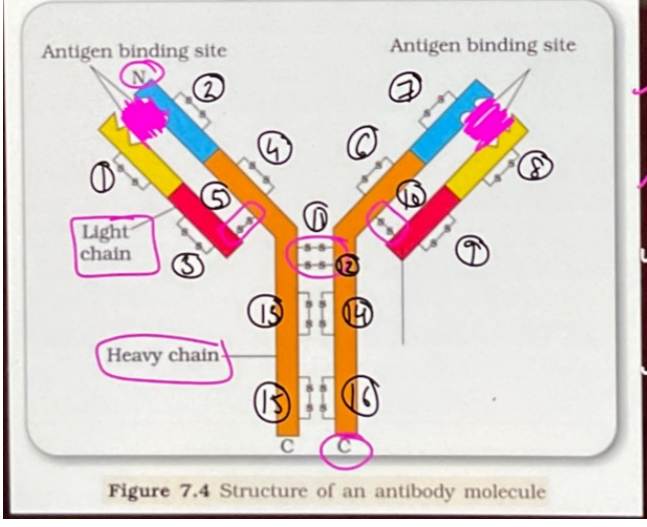


Figure 7.4 Structure of an antibody molecule

LC-HC → 1 Disulphide bond (each)

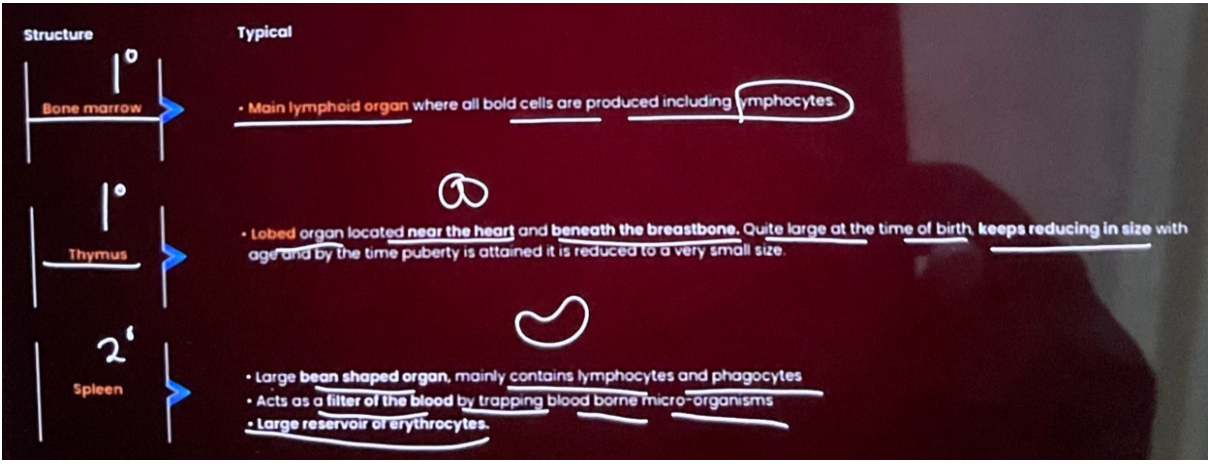
2HC - 2 Disulphide bond

16 Disulphide Linkage

2 antigen b.s

T lymphocytes are responsible for graft rejection. tissue and blood group matching are essential before undertaking any graft or transplant and even after this patient has to take immunosuppressants throughout life. If the pathogen succeed in gaining entry to our body, specific antibodies and T cells serve to kill these pathogens.

Lymphoid structures / Organs



2°

Lymph nodes

- Small solid structures located at different points along the lymphatic system • Serve to trap the microbes/antigens which happen to get into the lymph and tissue fluid. Antigens trapped in the lymph nodes are responsible for the **activation of lymphocytes** present there and cause the immune response.

MALT

- Mucosa-associated lymphoid tissue is located within the lining of major tracts like respiratory, digestive and urinogenital tracts
- Constitutes about **50%** of lymphoid tissue in human body.

Vaccination and Immunisation

Types of immunity

	Active	Passive
Antibodies	Produced within the host body	Ready-made/preformed antibodies are directly given
Time taken for full /effective	Longer	Shorter
Memory cells	✓	✗
Examples	• Natural infection → Antibody production in host Vaccination Deliberate injection of living/dead microbes/proteins	• Mother —Placenta→ Foetus • Mother —Colostrum (IgA)→ Infant e.g. Anti-venom

- The antibodies produced in the host body against antigens would neutralize the pathogenic agents during actual infection.

01

- The vaccines also generate **memory B-cells and memory T-cells** that recognise the pathogen quickly on subsequent exposure and overwhelm the invaders with a massive production of antibodies.

02

- **Recombinant DNA technology** has allowed the large scale production of antigenic polypeptides of pathogen in **bacteria/yeast**, hence greater availability for immunisation. e.g., **hepatitis B vaccine produced from yeast**

03

Allergies

Exaggerated response of immune system to certain antigens present in environment

Allergens are substances to which exaggerated immune response is produced example Pollens, mites in dust, animal dander etc
Antibodies- igE type

Symptoms- sneezing, watery eyes, running nose, difficulty in breathing
Chemical released- histamine and serotonin from mast cells
Diagnosis- patient is exposed to or injected with very small doses of possible allergens and reaction studied
Treatment- antihistamine, adrenaline and steroids quickly reduce the symptoms of allergy

Effects of modern day life style

- More and more children in metro cities of India suffer from allergies and asthma due to more sensitivity to the environment
- Protected environment provided early in life has resulted in lowering of immunity and person is more sensitive to allergens

Autoimmunity

Memory Based acquired immunity evolved in higher vertebrates can distinguish foreign molecules as well as foreign organisms from cells

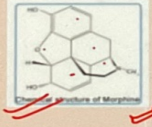
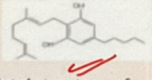
Results - self destruction/ body attack self cells.

reason - genetic / unknown

example — Rheumatoid arthritis.

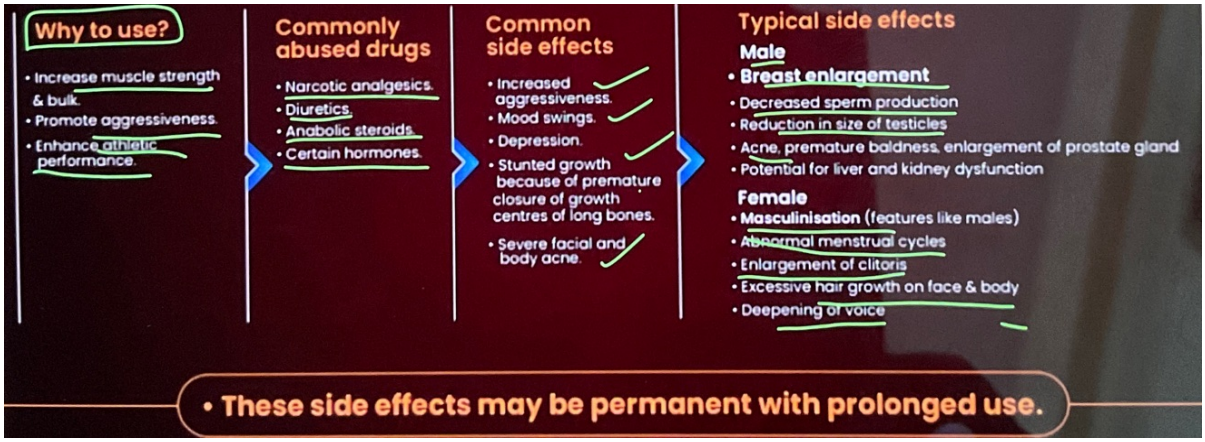
Drug Abuse

- Chemical when taken for a purpose other than medicinal use or in amounts/ frequency impairs one's physical, physiological or psychological functions and constitutes drug abuse.
- Source – Majorly from flowering plants and some from fungi.
- Commonly abused drugs are:

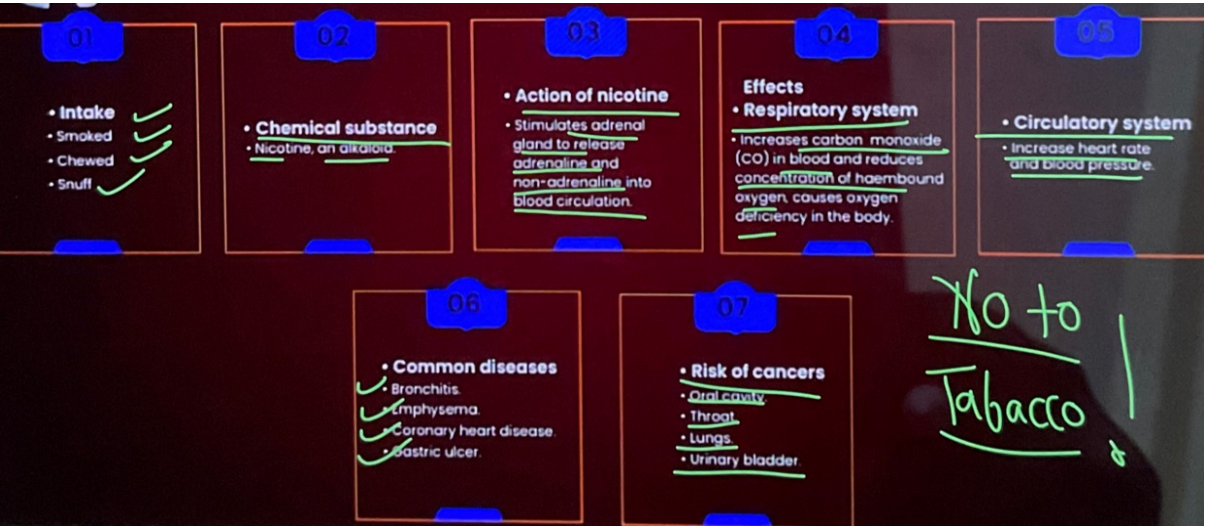
Drug	Receptors	Source	Intake	Examples	Action and anything specific
Opioids (1)	CNS, GIT	Latex of poppy plant, Papaver somniferum  Opium poppy	Snorting, injection 	• Morphine • Heroin/Smack (Diacetylmorphine)	• Effective sedative and pain killer • Useful in patients undergone Surgery • Depressant and slows down body functions • Odourless, white, bitter crystalline compound
Cannabinoids (2)	Principally in brain	Inflorescence, flower tops, leaves and resin of cannabis plant, Cannabis sativa  Leaves of Cannabis sativa	Inhalation, oral ingestion  Skeletal structure of cannabinoid molecule	• Charas • Hashish • Ganja • Marijuana	• Produced by virus infected cells that protect non-infected cells from further infection • Effects on cardiovascular system of the body • These days cannabinoids are formating also being abused by some sportspersons

Drug	Receptors	Source	Intake	Examples	Action and anything specific
Stimulants (3)	CNS	Coca plant Erythroxylum coca (Native of South America	Snorting	• Cocaine/coka alkaloid • Commonly called (coke/crack)	• Interferes with transport of neurotransmitter dopamine. • Potent stimulating action on CNS, producing sense of euphoria and increased energy • Excessive dosage causes hallucinations
Hallucinogens (4)		Atropa belladona, Datura	 Flowering branch of Datura		• Have been used for hundreds of years in folk-medicine, religious ceremonies and rituals all over the globe
Other drugs (5) Tranquilizers		Synthetic		Barbiturates, Benzodiazepines, Amphetamines	• Help patients cope with mental illness like depression insomnia.

Drugs and sportsperson



Tobacco/smoking – paves the way to hard drugs.



Adolescene and drug / Alcohol abuse

Adolescence means both a period and process during which a child mature in terms of his/her, attitudes and beliefs for effective participation in society

Adolescence is a bridge linking childhood and adulthood

It is a period between 12-18 years of age of vulnerable phase of mental and psychological development of an individual

It is accompanied by several biological and behavioural changes

Curiosity, need for Adventure and excitement and experimentation, motivate youngsters towards drugs and alcohol use

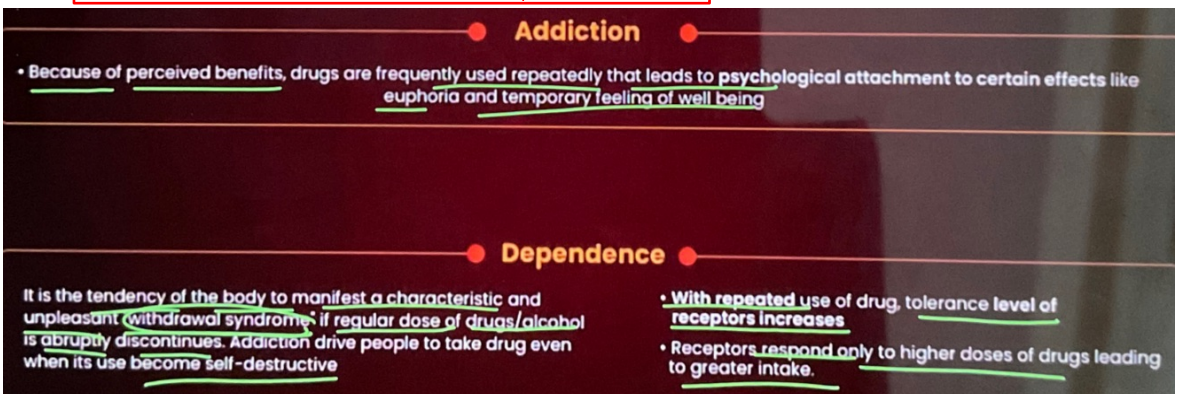
First use maybe out of curiosity, but later used to escape from stress, pressures to excel in academics. Perception that it is cool

Television movies, newspapers, internet promote this perception

Unstable or unsupportive, family structures and peer pressure also promote drug and alcohol abuse

Use of drugs, even once be fore- runner to addiction and pull the user into a vicious circle, leading to their regular use/ abuse

Addiction and Dependence



Effects of drug/alcohol abuse

• Reckless behavior. • Vandalism • Violence • Depression • Fatigue • Drop in academic performance

Warning signs

- Unexplained absence from school/college
- Aggressive and rebellious behaviour
- Change in sleeping and eating habits
- Deteriorating relationships with family and friends
- Poor personal hygiene, withdrawal, isolation
- Loss of interest in hobbies
- Fluctuations in weight and appetite
- High doses lead to coma and death due to respiratory failure, heart failure or cerebral hemorrhage
- Chronic use of drugs/alcohol damage nervous system and liver (cirrhosis)
- Use of drugs during pregnancy adversely affect foetus. **Some far-reaching implications**
- Abuser may turn to stealing
- Addict becomes the cause of mental and financial distress to entire family and friends

Withdrawl syndrome

If drug is abruptly discontinued, symptoms include:

• Anxiety • Nausea • Shakiness • Sweating

• In severe cases, can be life threatening, person needs a medical supervision.

Prevention and control

"Prevention is better than cure"

- **Avoid undue peer pressure** on child related to sudies, sports or other activities
- **Education and counselling:** Channelise energy of child into healthy pursuits like sports, yoga, reading, music, etc.
- Sort out problems by **seeking help from parents and peers.**
- **Looking for danger signs :** Alert parents, teachers and close friends need to look for and identify the danger signs of substance (drug/alcohol) abuse and appropriate measures would then be required to diagnose the malady and underlying cause.
- Proper remedial steps or treatment should be taken by **seeking professional and medical help** in the form of highly qualified psychologists, psychiatrists and de-addiction and rehabilitation programmes. This will totally relieve the individual from these evils.

THANK YOU!!